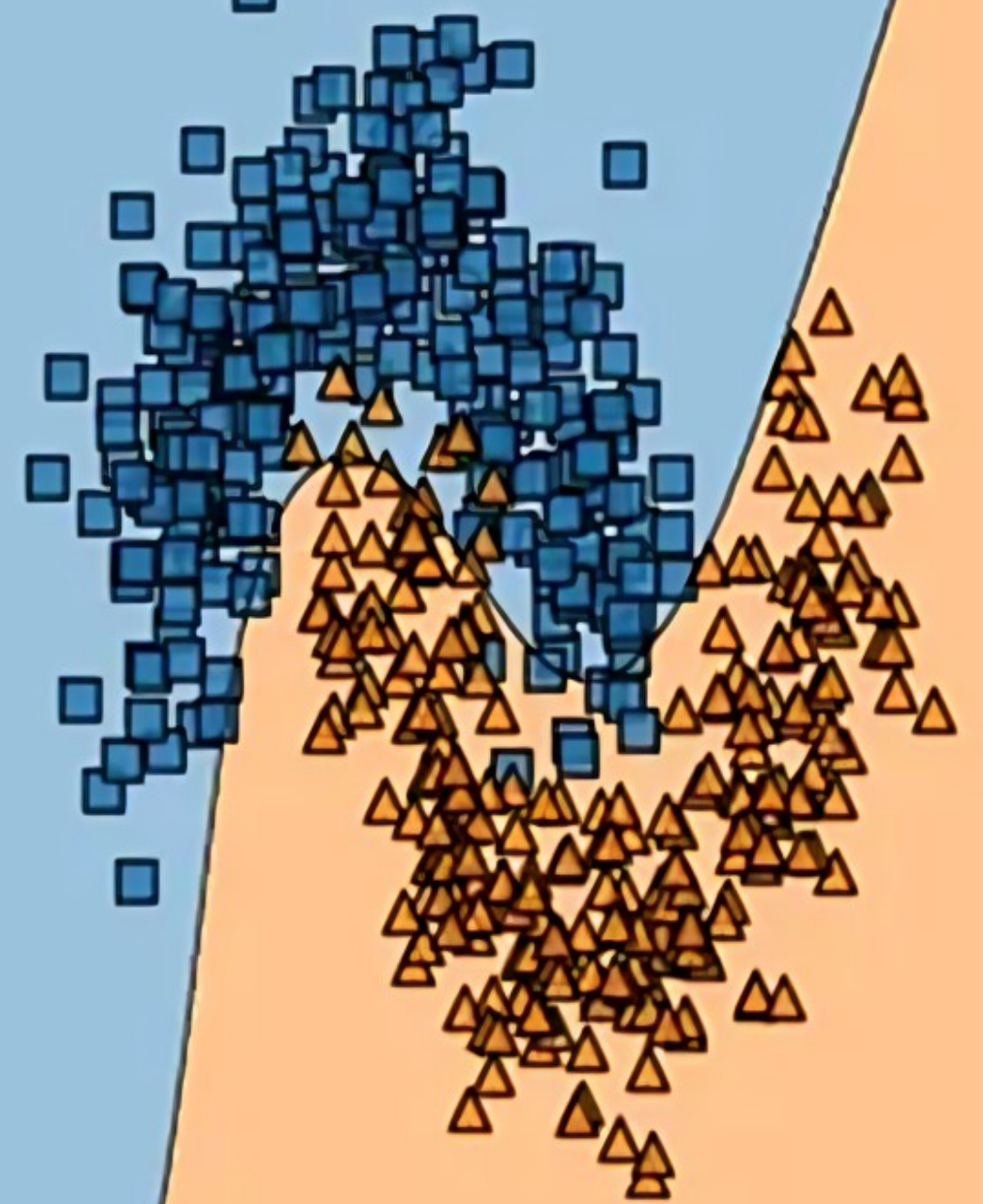




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SigmaCam: Exact Decision Boundary Extraction for DNNs with Smooth Nonlinearities

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The Challenge of Neural Network Interpretability



Black Box Problem

Neural networks are often considered "black boxes" due to the complexity of their internal decision-making mechanisms, leading to challenges in explaining their predictions.



Visualization Need

Visualization of decision boundaries provides a powerful tool for interpreting the behavior of neural networks and understanding how they partition input space for classification.

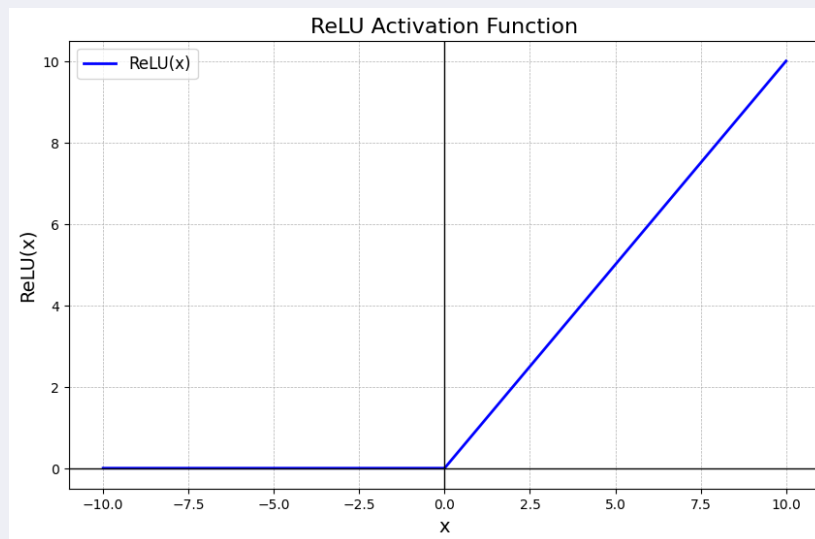


Smooth Activation Challenge

State-of-the-art models predominantly utilize smooth nonlinear activations such as Sigmoid and SiLU, which pose additional challenges for decision boundary extraction due to their continuous and differentiable nature.



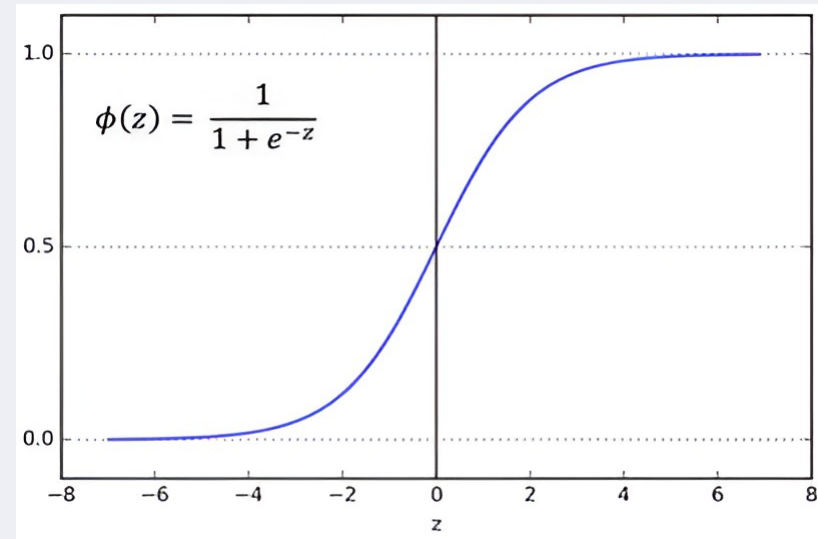
Key Activation Functions in Our Study



ReLU Activation

$$\text{ReLU}(x) = \max(0, x) \quad - (1)$$

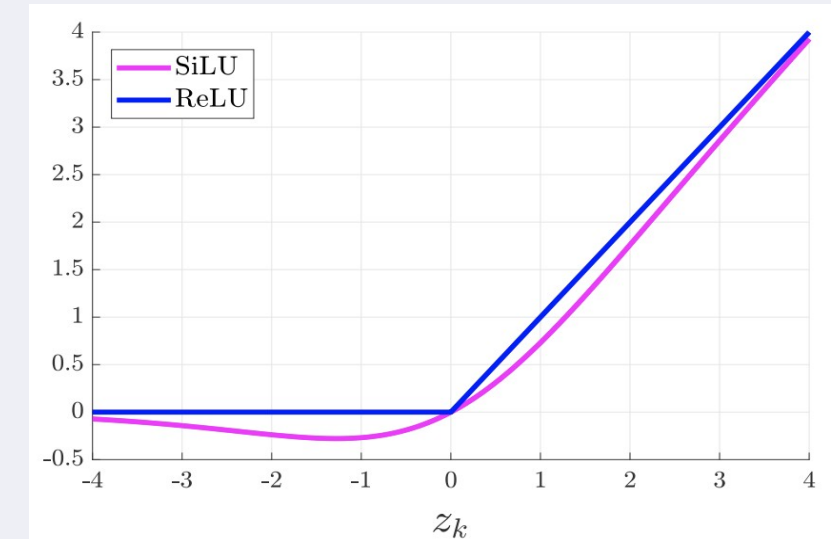
ReLU-activated networks produce piecewise-linear decision boundaries with more jagged contour lines, creating sharper transitions between classes.



Sigmoid Activation

$$\text{Sigmoid} = \sigma(x) = 1/(1 + e^{(-x)}) \quad - (2)$$

Sigmoid-activated MLPs generate smooth but distinct class separations, providing clearer visualization of the decision boundary.



SiLU Activation

$$\text{SiLU}(x) = x \cdot \sigma(x) = x/(1 + e^{(-x)}) \quad - (3)$$

SiLU-activated networks demonstrate more natural and fluid transitions between classes, with boundaries that more elegantly align with the underlying data geometry.

SigmaCam: Methodology Overview

Input Domain Initialization

Initialize the high-dimensional input domain for analysis, setting the foundation for boundary extraction.

2D Projection & Grid Generation

Project onto a two-dimensional plane using a projection matrix and generate a structured grid within this 2D space.

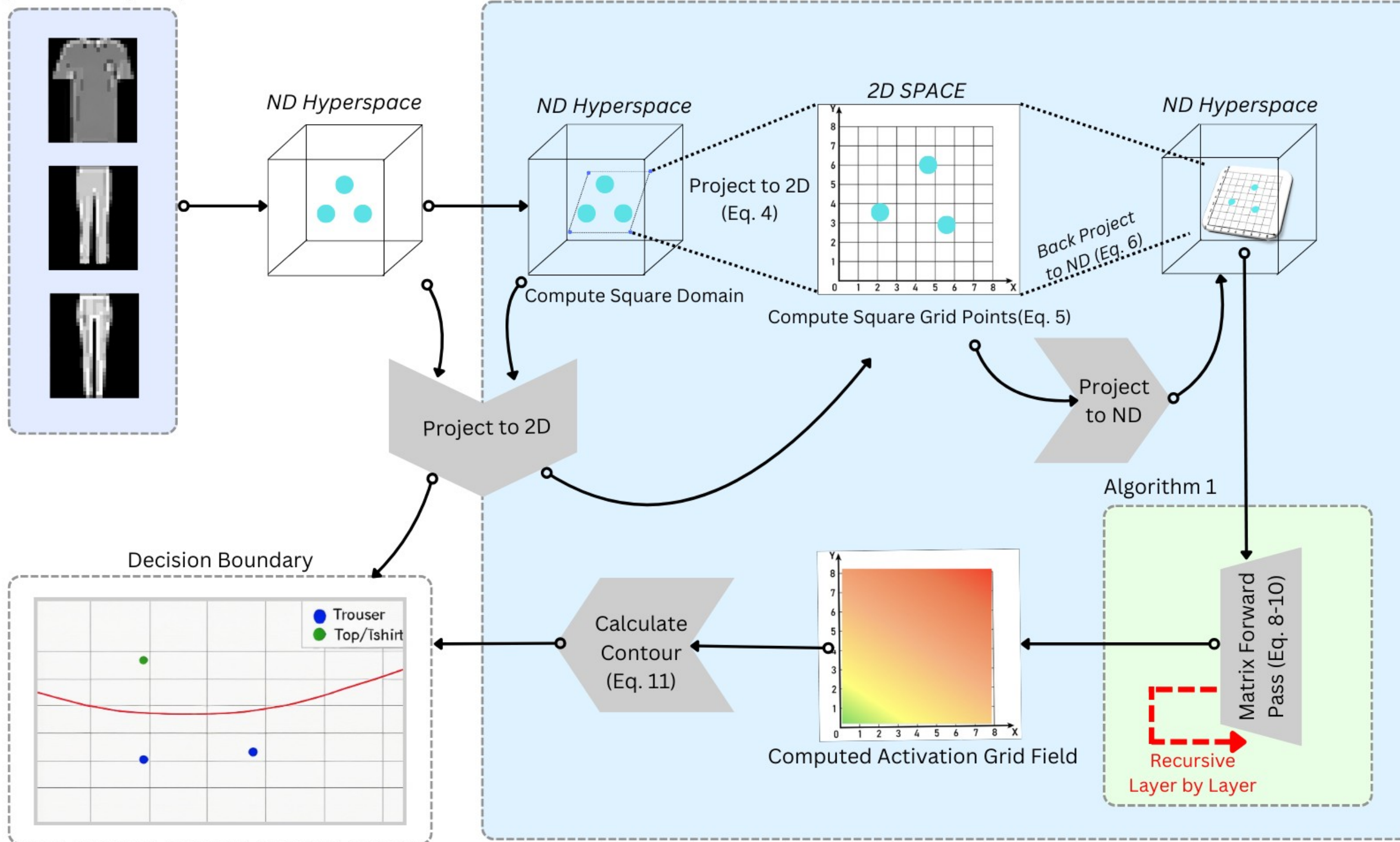
Matrix-Based Forward Pass

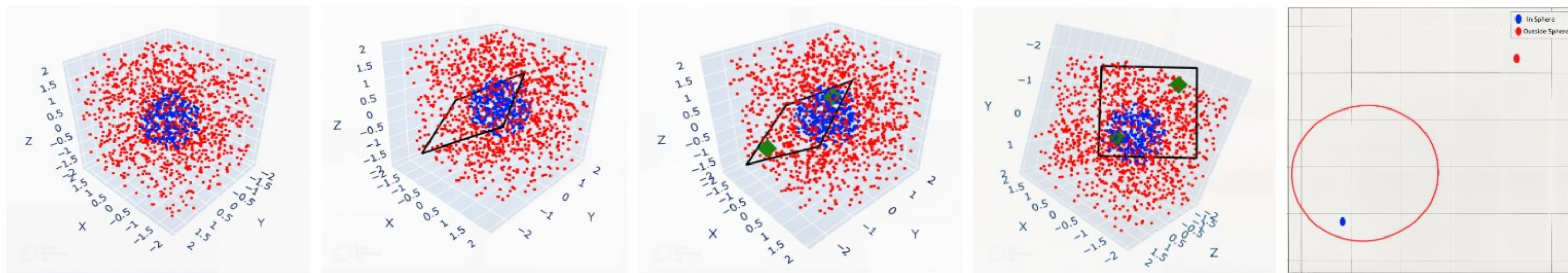
Compute activations recursively at each layer through a matrix-based forward pass of the trained MLP.

Decision Boundary Evaluation

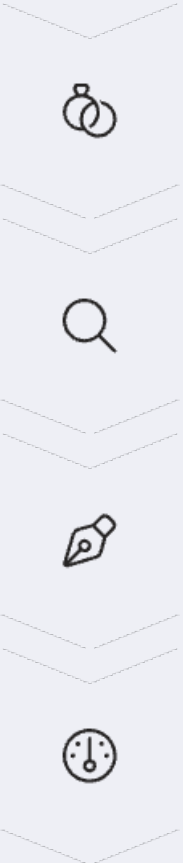
Evaluate final activations against a predefined threshold to delineate the exact decision boundary.

Samples From Input Space





Recursive Decision Boundary Computation



Matrix Forward Pass

Begin with an input grid of candidate points and compute activations at each layer via a matrix forward pass: $z^{(l)} = \sigma(W^{(l)}z^{(l-1)} + b^{(l)})$ - (4)

Recursive Activation Computation

Recursively compute activations for each subsequent layer until reaching the final output layer, then compare against threshold τ

Contour Extraction

Generate a two-dimensional projection of the input domain to visualize the decision boundary as a contour

Efficiency Optimization

Utilize matrix operations and mesh grids with GPU acceleration to enhance computational performance

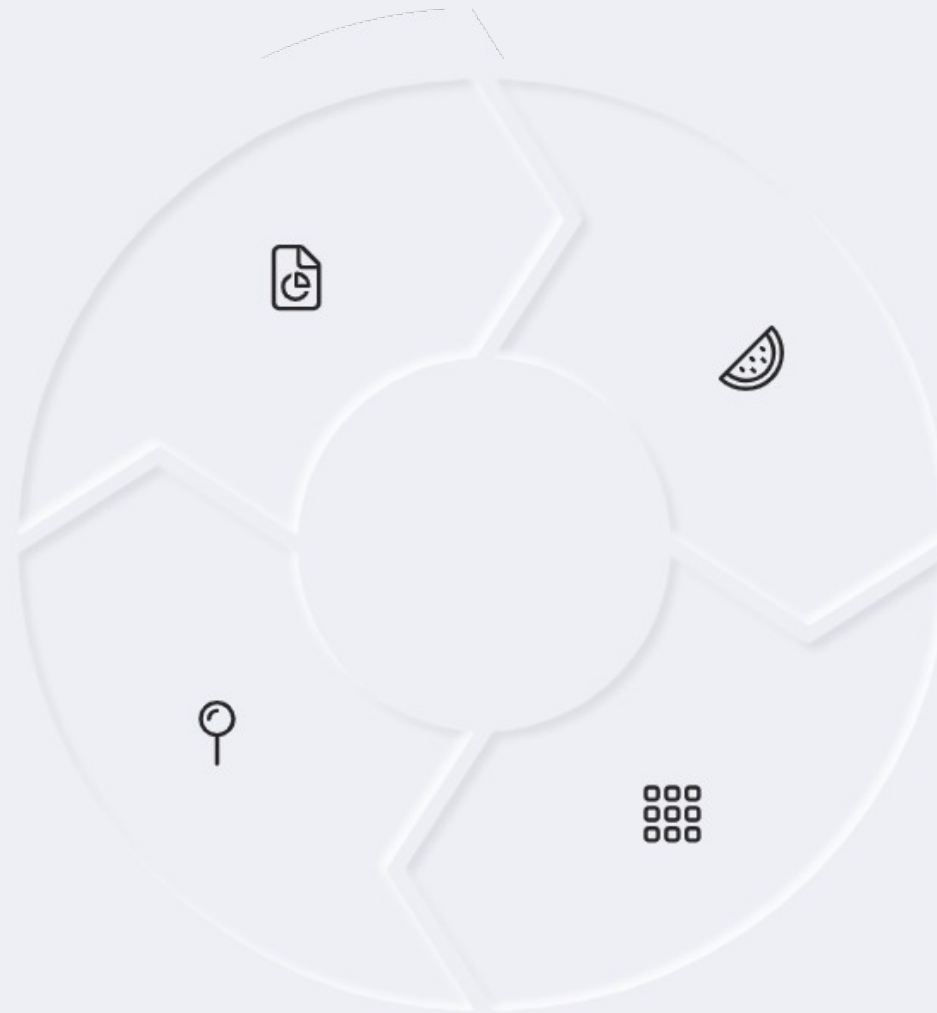
Handling Higher Dimensions

Projection Techniques

Project high-dimensional input spaces onto lower-dimensional manifolds for visualization

Manifold Mapping

Map high-dimensional decision boundaries onto interpretable two-dimensional planes



Slicing Methods

Slice high-dimensional spaces into two-dimensional sections for boundary analysis

Structured Grid Generation

Generate structured grids in 2D space and back-project to original dimensions

Algorithm Implementation

Algorithm 1: Matrix-Based Decision Boundary Extraction

Takes a trained MLP model with weights and biases, activation function, grid points, and layer index as inputs. Recursively computes activations through the network layers until reaching the final layer, returning decision boundary values for visualization.

Algorithm 2: 2D Decision Boundary Visualization

Takes an input domain, projection matrix, trained model, and threshold as inputs. Computes projection matrix and centroid, projects domain to 2D, creates a grid, back-projects to original space, evaluates the model, and plots the contour at the threshold value.

Computational Efficiency

By circumventing analytical inversions, the method achieves both scalability and precision. GPU acceleration ensures practical computational performance, making the algorithm suitable for complex and high-dimensional models.

Experimental Setup



Framework

PyTorch implementation with MLPs (2-5 layers) and input dimensions ranging from 2D for direct visualization to higher dimensions with projected slices



Activation Functions

Sigmoid and SiLU (Swish), as well as their mixtures, confirming applicability beyond piecewise polynomial activations



Datasets

Synthetic datasets (spheres, spirals) and real-world datasets (MNIST, Fashion-MNIST, MedMNIST) reduced to binary tasks and projected into 2D domains

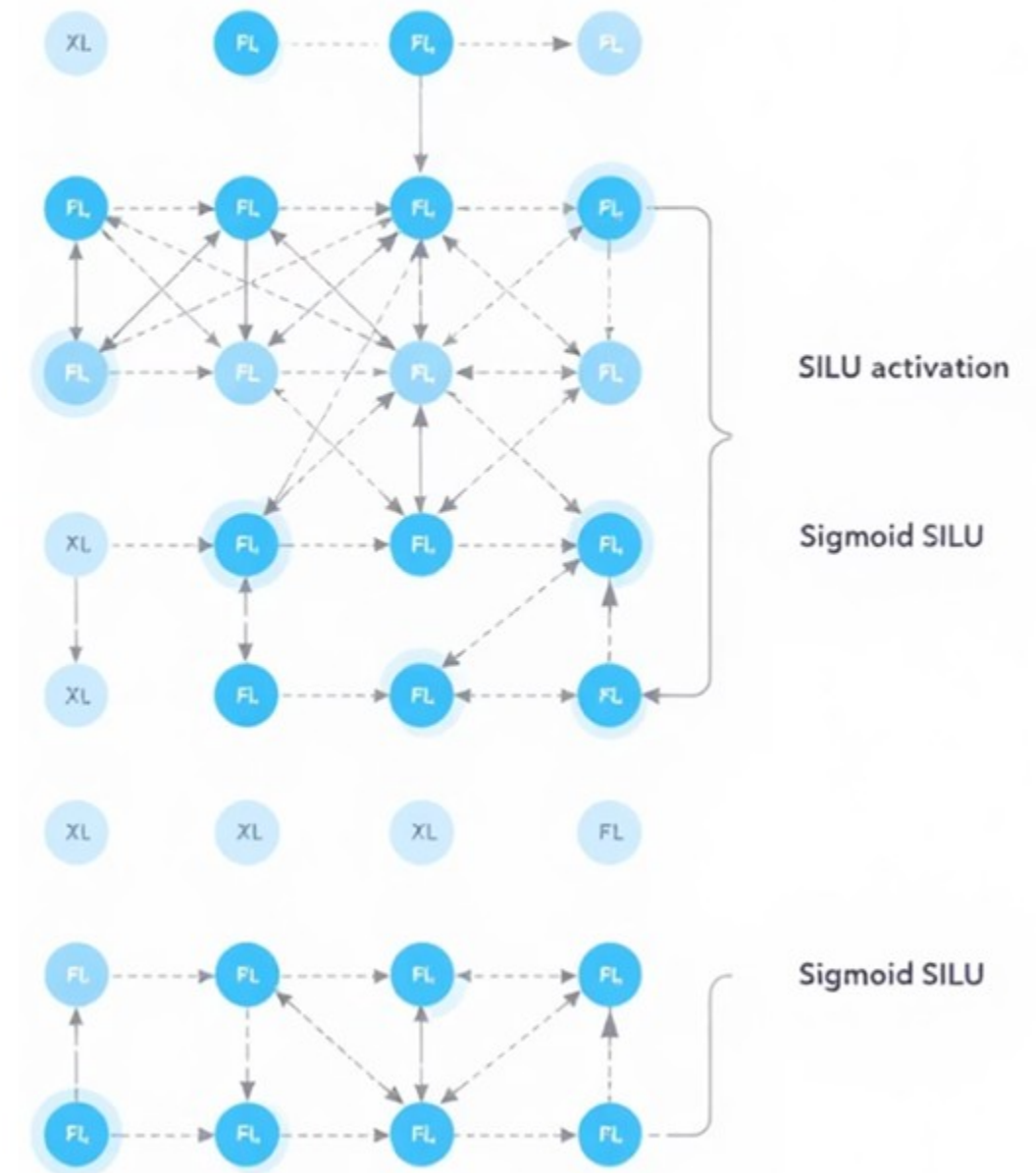


Hardware

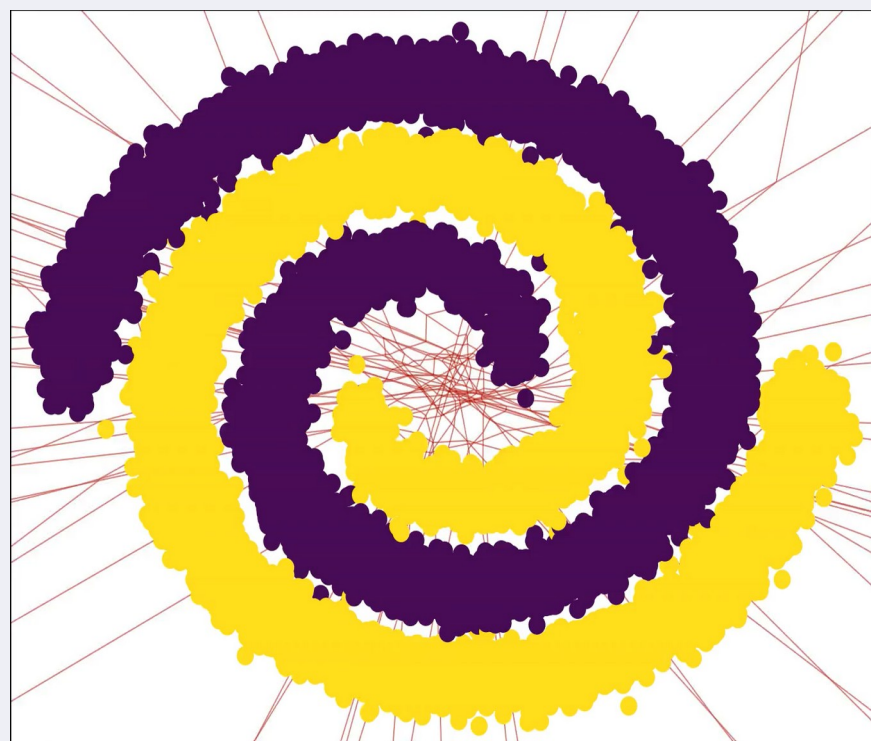
All experiments run on commodity hardware (T4 GPU)

Neural Network Signitics

Neural network sigmodic and sigmoid and SILU activation functions

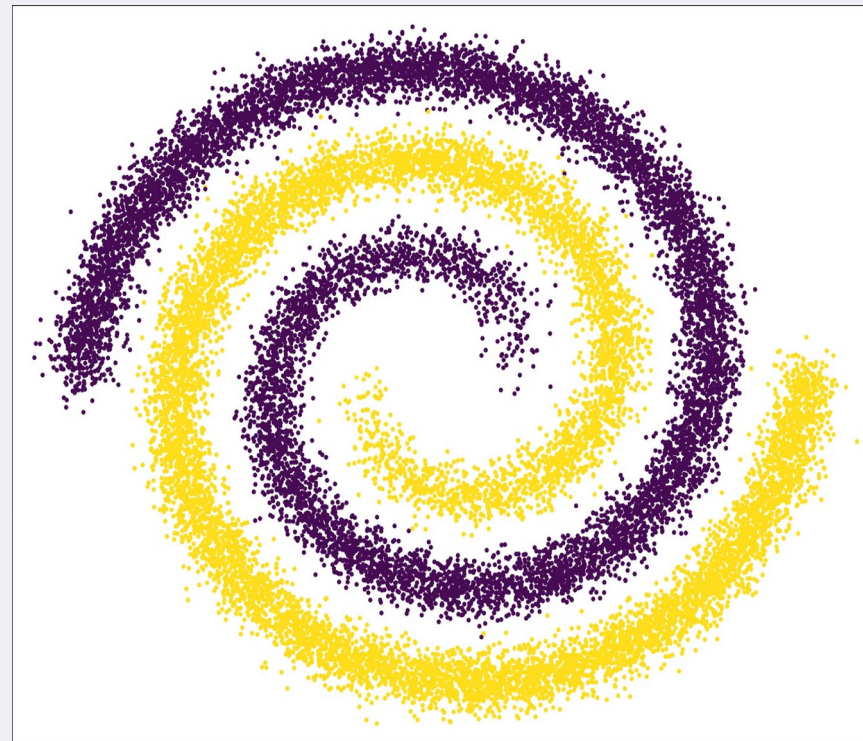


Spiral Dataset Results



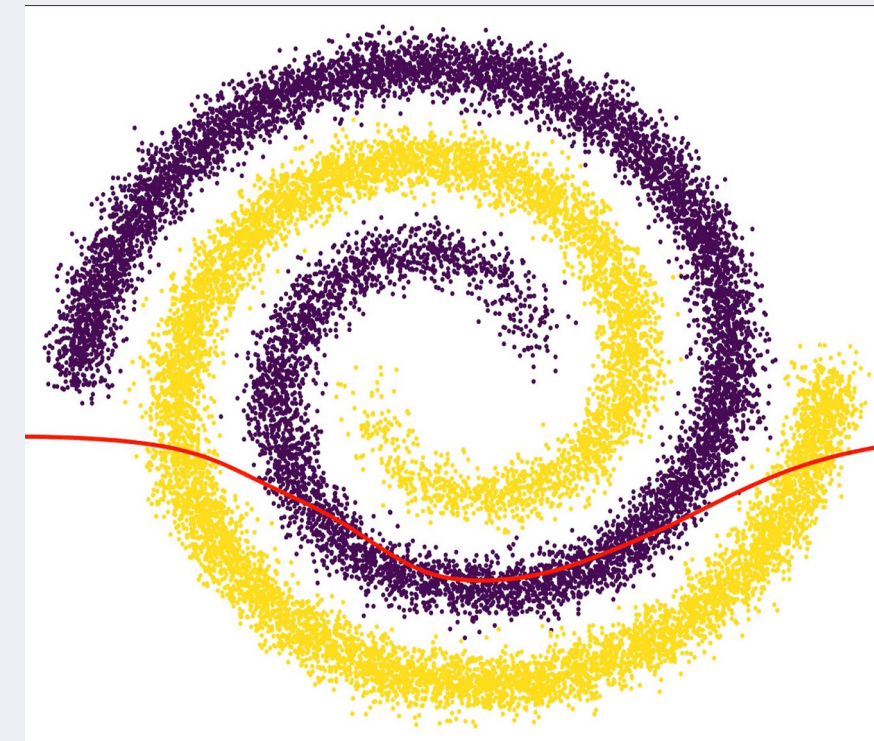
ReLU (SplineCam)

SplineCam on a ReLU-activated MLP produces exact, piecewise-linear contours that closely follow most spiral turns but exhibit sharp kinks at activation hinges, highlighting how CPWL activations struggle to model smooth, highly nonlinear boundaries .



Sigmoid (SigmaCam)

SigmaCam extends exact boundary extraction to Sigmoid-activated networks, generating smooth, continuous contours that gracefully trace every winding of the spiral arms, demonstrating our recursive algorithm's ability to handle differentiable activations without approximation



SiLU (SigmaCam)

With SiLU (Swish) activations, SigmaCam yields even more natural and fluid decision frontiers that align seamlessly with the spiral geometry, showcasing improved gradient propagation and boundary fidelity compared to both ReLU and Sigmoid cases

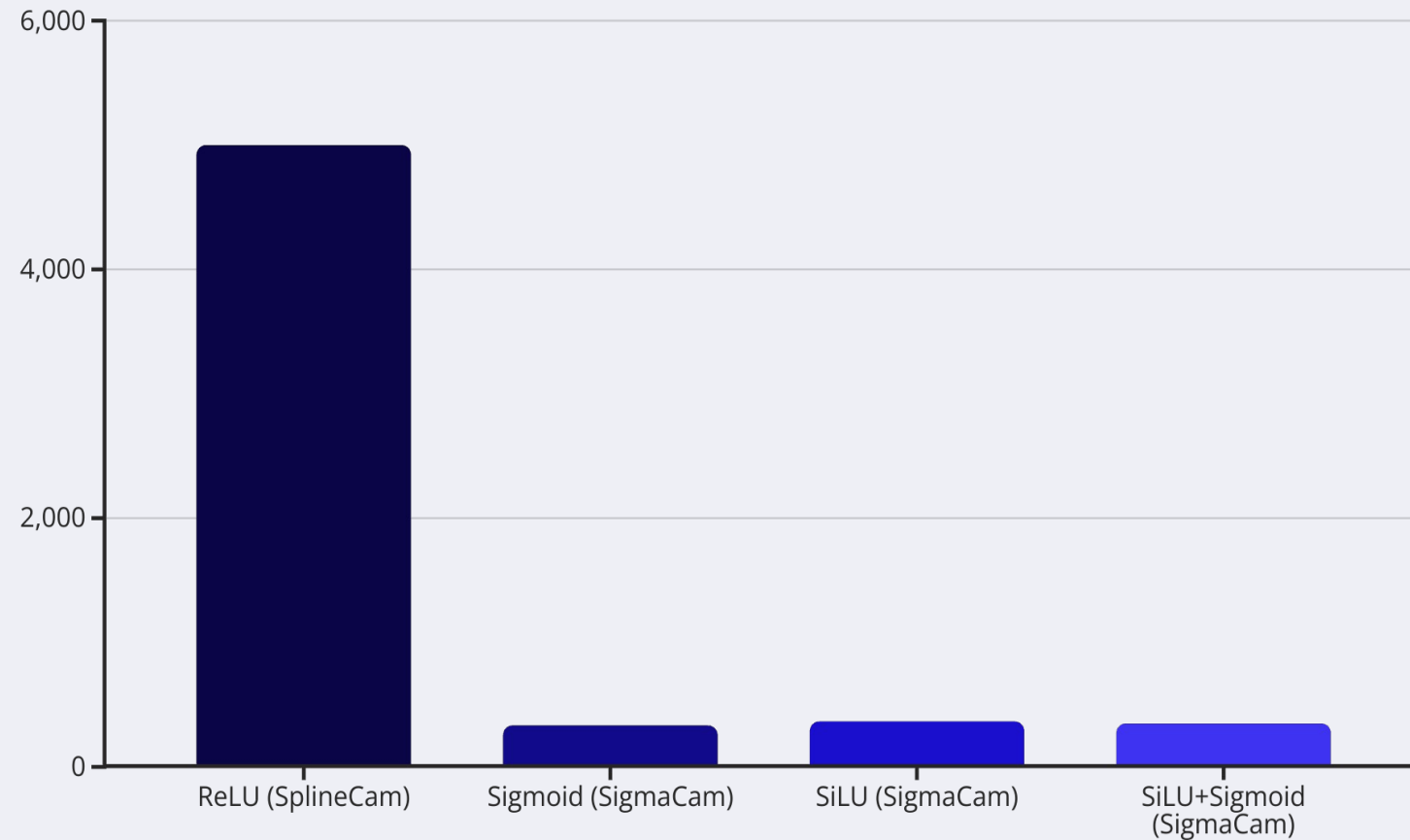
Spiral Dataset Results

Metric	Sigma Cam		Spline Cam
	SiLU	Sigmoid	ReLU
Accuracy	0.99825	0.98395	0.99675
Precision	0.999198	0.989184	0.998201
Recall (Sensitivity)	0.9973	0.9786	0.9993
F1 Score	0.998248	0.983864	0.99875
R^2 Score	0.993	0.9358	0.995
Smoothness	668.5279	494.4718	768.8812
Gradient Magnitude	0.399401	28.78554	1.0528

Spiral Dataset Results

EPOCH	SIGMOID		SILU		RELU	
	ACCURACY	TIME ELAPSED (SECONDS)	ACCURACY	TIME ELAPSED (SECONDS)	ACCURACY	TIME ELAPSED (SECONDS)
50	0.73625	80.47	0.9984	100.03	0.99825	866.12
100	0.79905	80.59	0.99875	85.37	0.99845	1098.87
150	0.92275	81.39	0.9988	86.43	0.99865	1368.40
200	0.97985	82.14	0.9988	84.64	0.9985	1666.79
Total time		324.59		356.47		5000.18

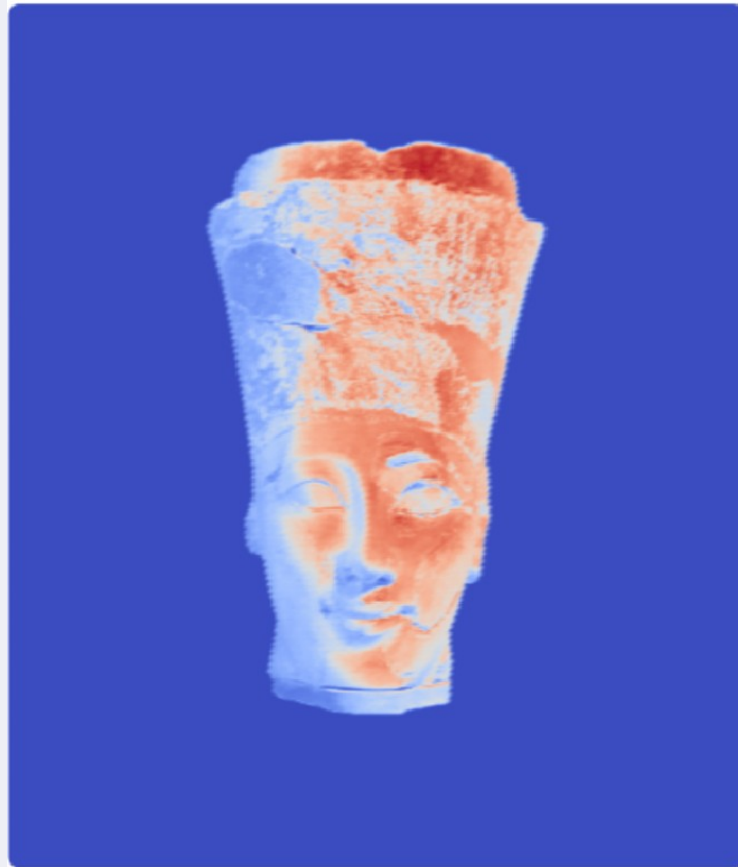
Computational Performance



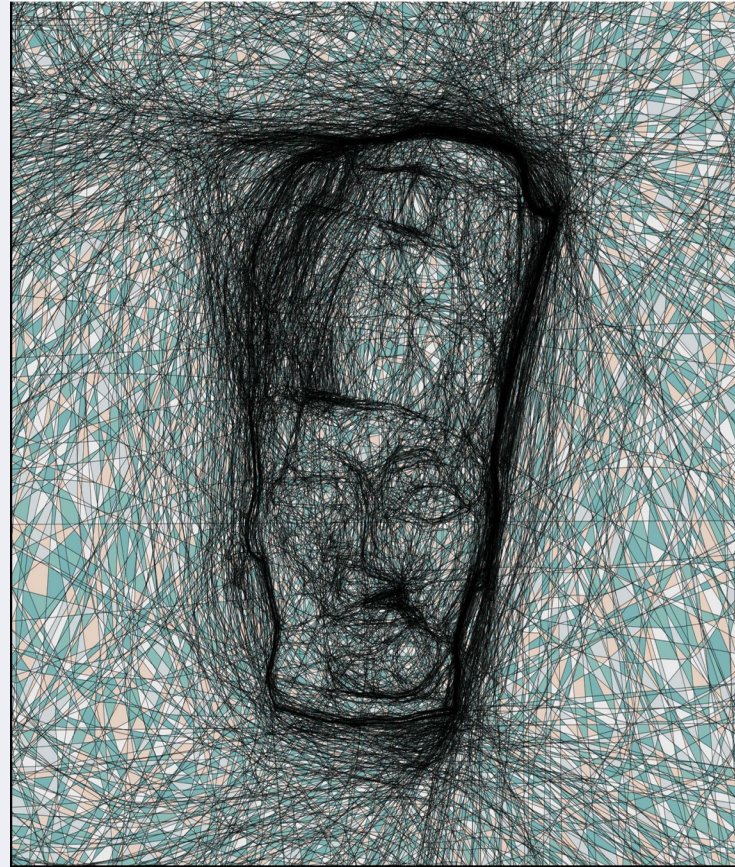
SigmaCam demonstrates superior efficiency when utilizing smooth activation functions such as Sigmoid and SiLU, achieving high accuracy with significantly lower computation times compared to SplineCam with ReLU activation.

While SplineCam requires 5,000 seconds total computation time, SigmaCam completes the task in just 325-356 seconds across all configurations, maintaining similar or better accuracy levels. The SiLU+Sigmoid model achieved the highest accuracy while maintaining competitive execution times.

Facial Image Embeddings



Raw facial-image embeddings in 2D latent space, revealing the underlying class structure.



SplineCam (ReLU) extracts exact piecewise-linear boundaries, but shows sharp, angular transitions.



SigmaCam (SiLU) traces smooth, natural contours that faithfully follow the complex class separations.

Facial Image Embeddings



Raw facial-image embeddings in 2D latent space, revealing the underlying class structure.



SplineCam (ReLU) extracts exact piecewise-linear boundaries, but shows sharp, angular transitions.



SigmaCam (SiLU) traces smooth, natural contours that faithfully follow the complex class separations.

MNIST Digits Results

Dataset Preparation

MNIST provides 70,000 grayscale 28×28 images of handwritten digits (0–9); we filtered to digits “2” (label 1) and “3” (label 0), split into 60 000 training and 10 000 test samples, then reduced to 2D via PCA or autoencoder embeddings

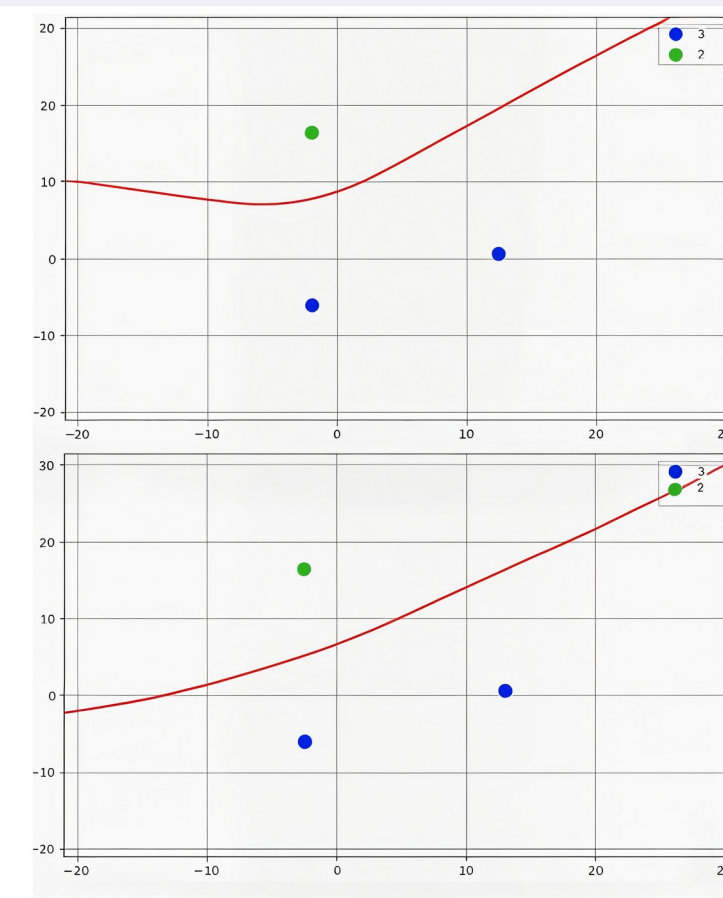
Model Training

A compact MLP with SiLU activations was trained on these 2D embeddings to distinguish between “2” and “3,” achieving high classification performance prior to boundary extraction

Boundary Visualization

SigmaCam was applied to the 2D embedding space, overlaying exact contour lines at the decision threshold to pinpoint the precise boundary separating the two digit clusters

MNIST Digit Classification



Using our recursive SigmaCam algorithm, we overlay exact red contour lines at the $f(x)=0.5$, threshold onto the blue/green 2D embeddings of digits “2” and “3,” precisely delineating how the network exploits subtle shape variations in latent space to distinguish these visually similar classes

Fashion MNIST Results

Dataset Preparation

FashionMNIST dataset filtered to focus on binary classification between "Trousers" (label 1) and "Tops" (label 0), containing 12,000 training samples and 2,000 test samples after filtering.

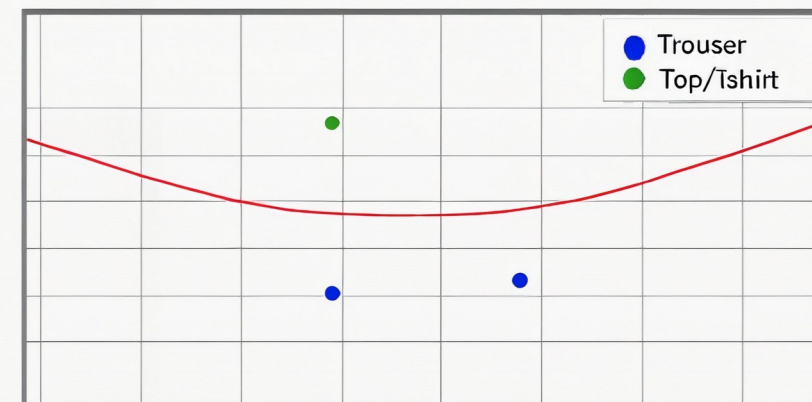
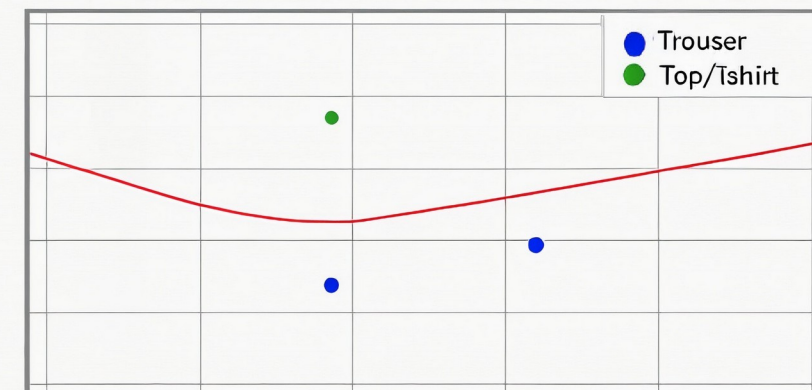
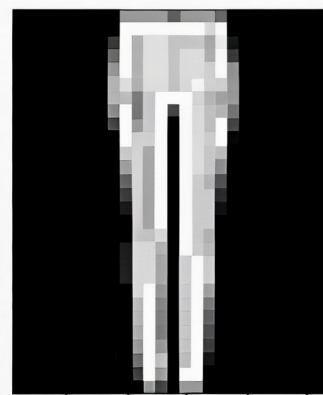
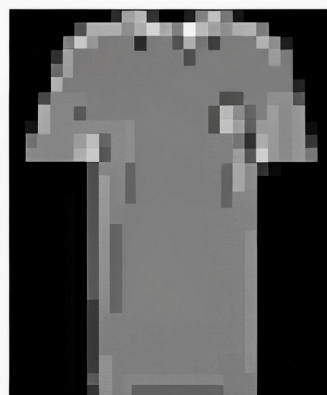
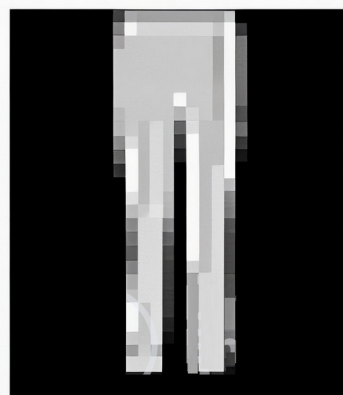
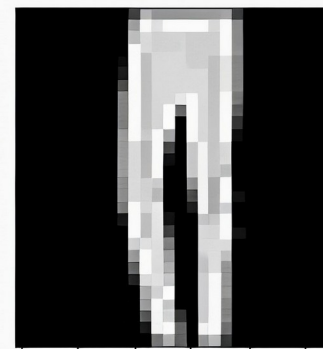
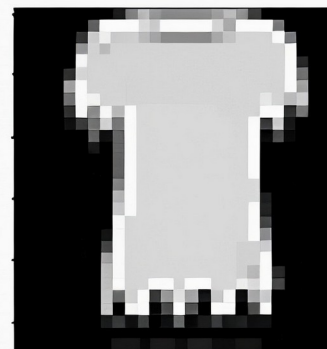
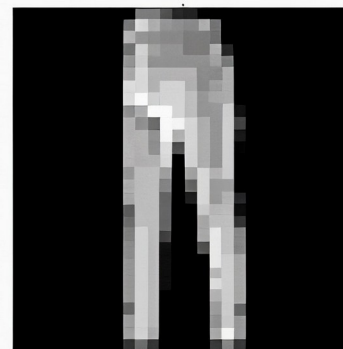
Model Training

Images normalized and relabeled for binary classification, with a small MLP using SiLU activations trained on the dataset to distinguish between the two visually distinct clothing categories.

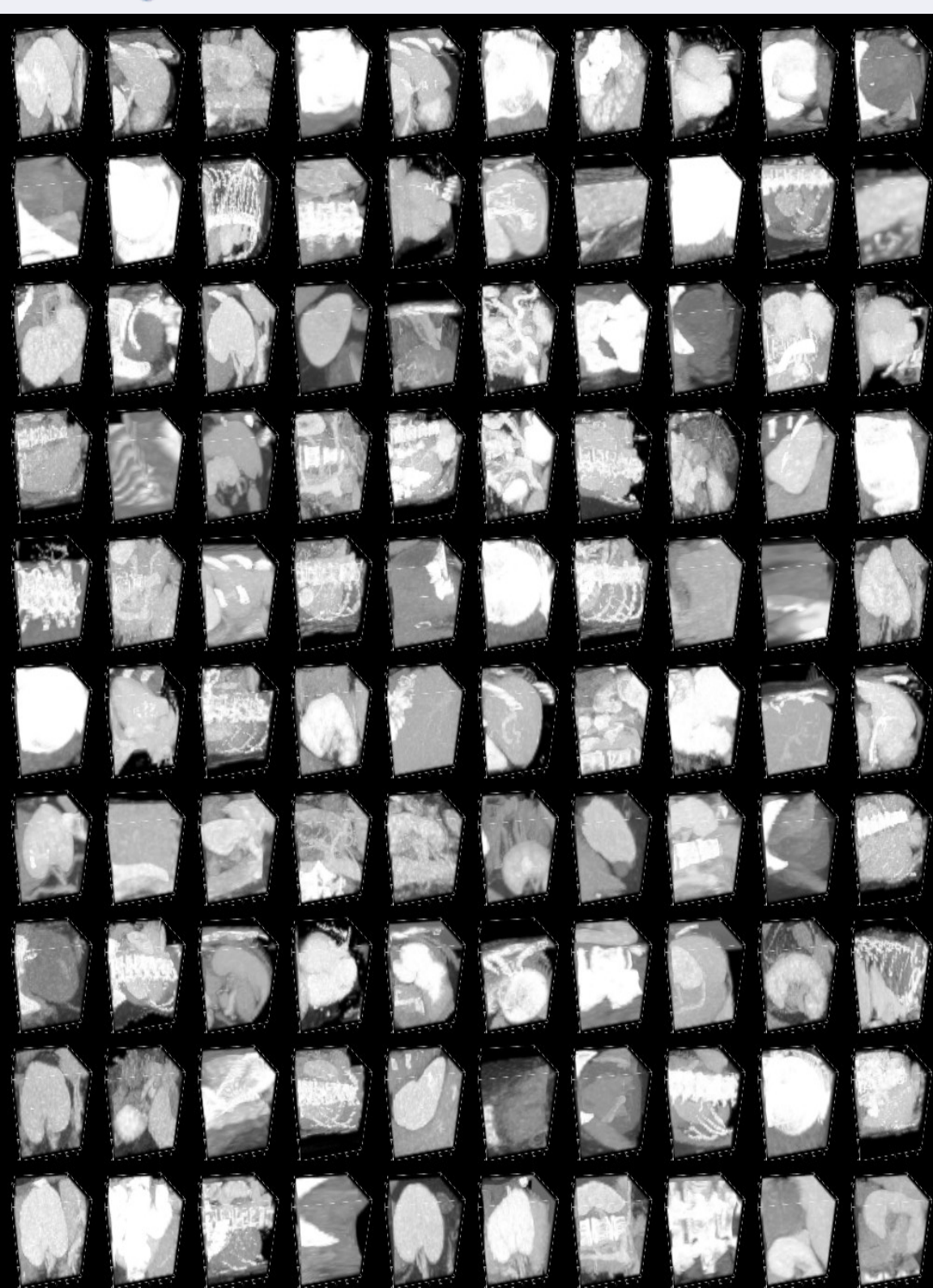
Boundary Visualization

SigmaCam applied to visualize decision boundaries in the input space, revealing where the model distinguishes between apparel categories with contour lines closely matching intuitive separations.

MNIST Digit Classification



Filtered to “Trousers” vs. “Tops,” we trained a compact SiLU-activated MLP and applied SigmaCam to overlay exact decision-threshold contours on sample images revealing smooth, shape-consistent boundaries directly in pixel space.



Medical Image Analysis



PneumoniaMNIST

5,856 grayscale chest X-ray images for distinguishing between normal and pneumonia-affected cases, with 4,708 training samples, 524 validation samples, and 624 test samples.



BreastMNIST

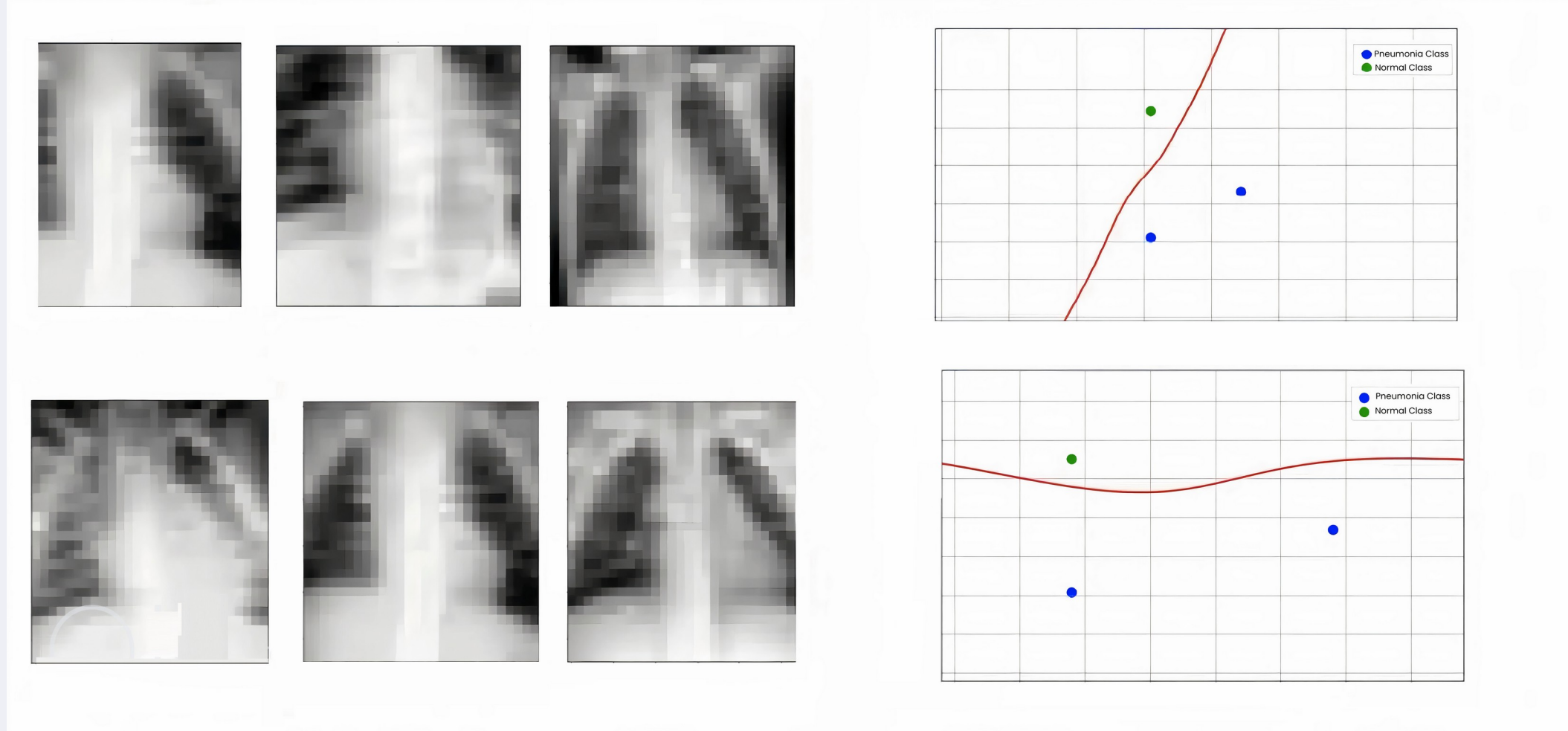
780 grayscale breast ultrasound images for classifying benign and malignant lesions, with a standard train-validation-test split of 546, 78, and 156 samples, respectively.



Visualization

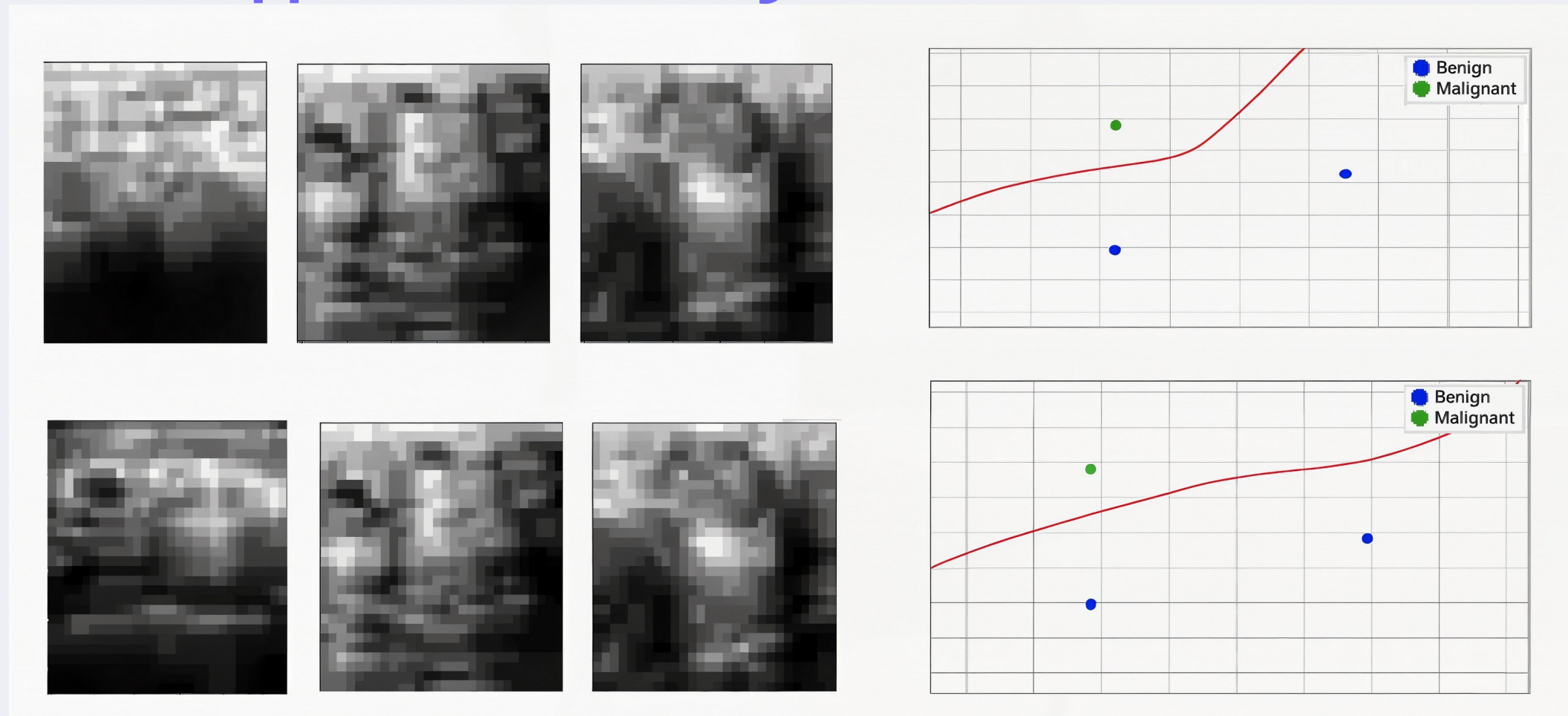
Decision boundaries extracted in 2D embeddings reveal how the model distinguishes between clinically relevant patterns, providing interpretable insights for medical professionals.

Real-World Applications of Image Classification: PneumoniaMNIST



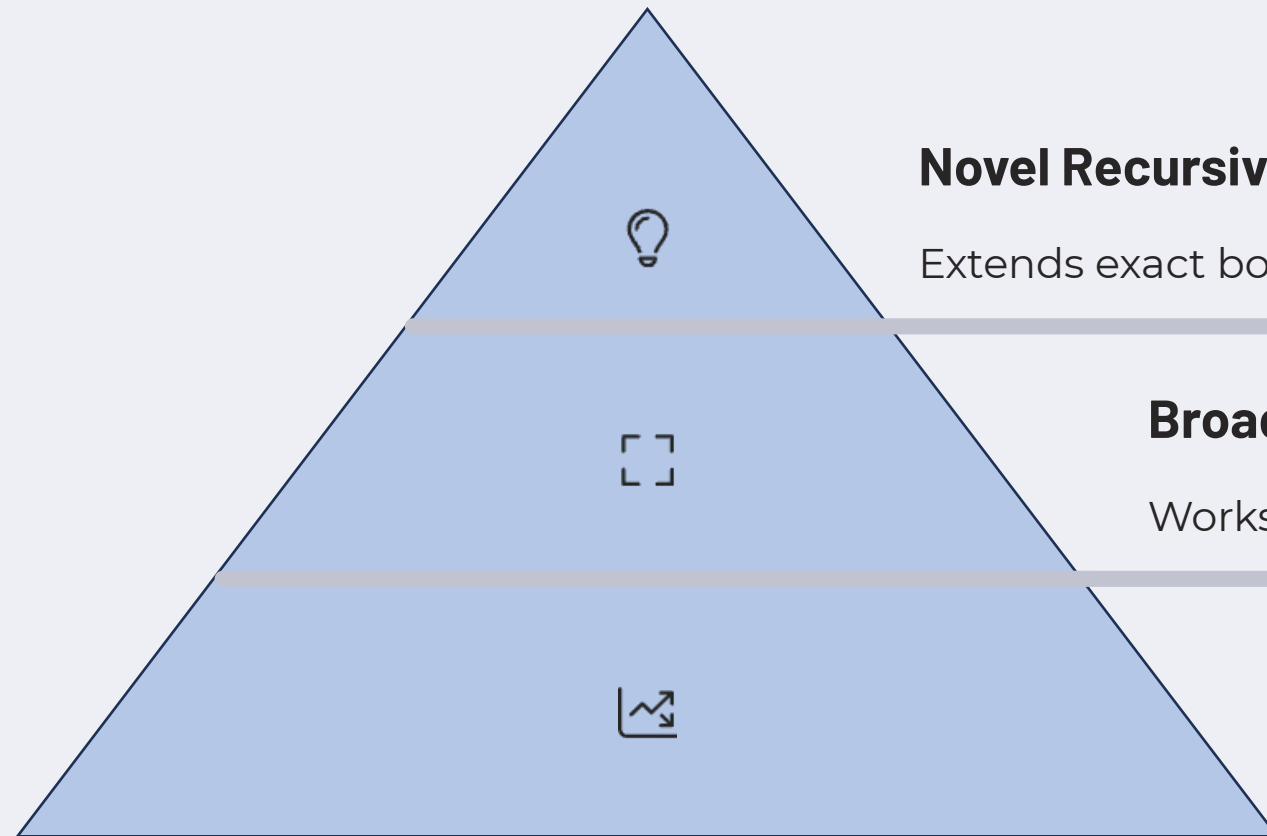
Filtered to “Pneumonia” vs. “Normal,” we trained a compact SiLU-activated MLP on chest X-rays and applied SigmaCam to overlay exact decision-threshold contours highlighting the subtle texture and intensity boundaries that distinguish healthy lungs from pneumonia.

Real-World Applications of Image Classification: BreastMNIST



Filtered to “Benign” vs. “Malignant,” we trained a compact SiLU-activated MLP on the 28×28 breast-tissue images and applied SigmaCam to overlay exact decision-threshold contours revealing the subtle intensity and texture cues that distinguish malignant lesions from benign tissue.

Conclusion and Future Work



Novel Recursive Algorithm

Extends exact boundary visualization to smooth activations

Broader Applicability

Works with Sigmoid, SiLU, and mixed activation functions

Computational Efficiency

15x faster than existing methods for exact boundary extraction

SigmaCam fills a significant gap by enabling exact decision boundary visualization for neural networks with smooth nonlinearities. Future work includes extending the method to larger architectures, investigating efficient parallelization strategies, and developing hybrid frameworks that integrate visual analytics with mathematical rigor.

By providing insights into model behavior through precise boundary visualization, SigmaCam enhances interpretability and trust in AI systems across diverse applications including medical imaging and computer vision.



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